

ECE461 - HW09

1)

$$y[m] = x[m] + \frac{1}{4} x[m-1] + w[m], \quad m \geq 1$$

1.a) With $N=2$, $H[k] = \sum_{n=0}^{N-1} h[n] e^{-j \frac{2\pi}{N} nk}$ end $h[0] = 1$
 $h[1] = 0.25$

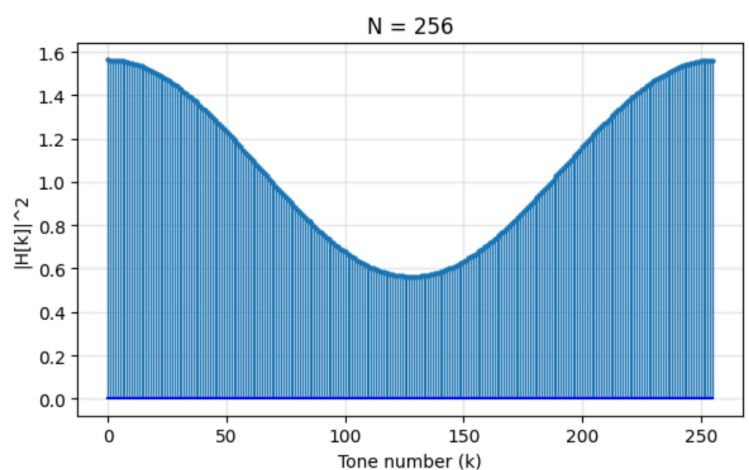
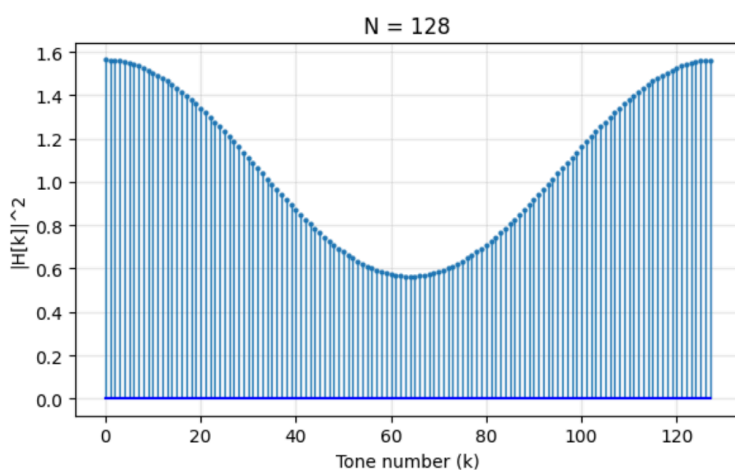
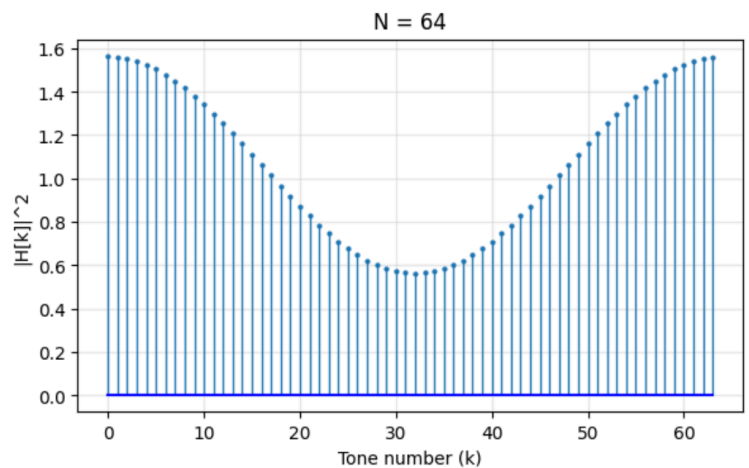
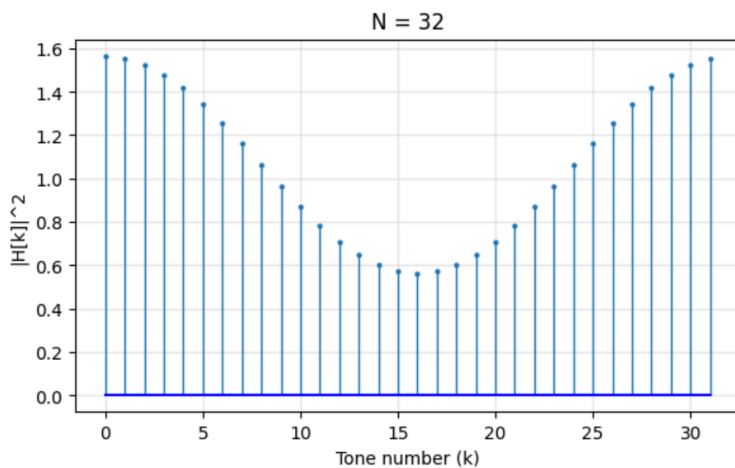
$$H[0] = h[0] e^0 + h[1] e^0 = 1 + 0.25 = 1.25$$

$$H[1] = h[0] e^0 + h[1] e^{-j \frac{2\pi}{2} 1 \cdot 1} = 1 - 0.25 = 0.75 //$$

1.b) We basically need to take an N -point DFT of $h[m]$

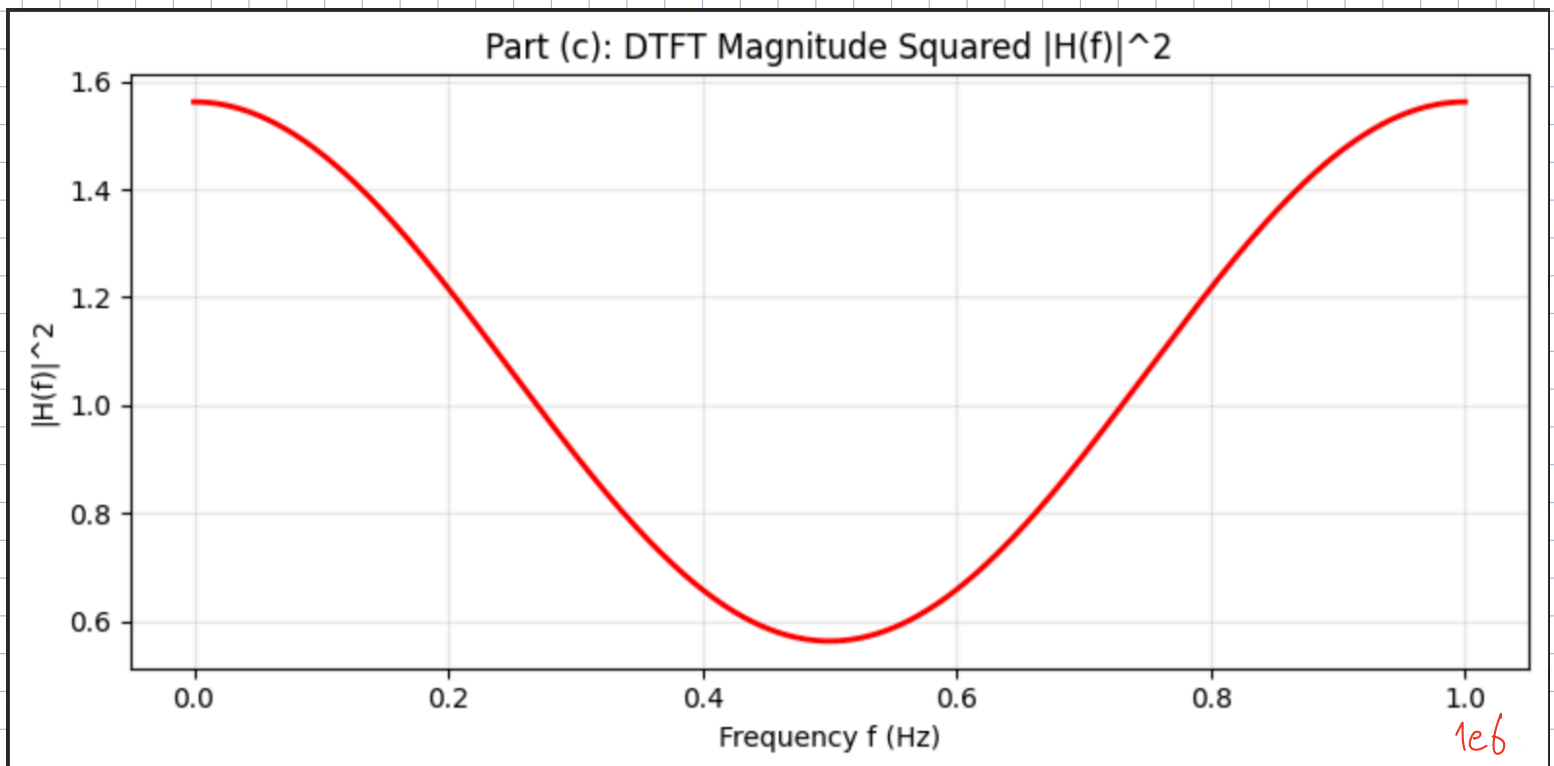
$$H[k] = h[0] + h[1] e^{-j \frac{2\pi}{N} k} = 1 + 0.25 e^{-j \frac{2\pi}{N} k}$$

$$|H[k]|^2 = 1.0625 + 0.5 \cos\left(\frac{2\pi}{N} k\right)$$

Part (b): $|H[k]|^2$ for various N 

$$1.c) \quad H(f) = \sum_{l=0}^{L-1} h_l \exp\left(-j \frac{2\pi f l}{W}\right), \quad f \in [0, W]$$

$$= h_0 + 0.25 h_1 e^{-j \frac{2\pi}{W} f} \Rightarrow |H(f)|^2 = 1.0625 + 0.5 \cos\left(\frac{2\pi}{W} f\right)$$



Notice that over one period N -point DFT coefficients found in part b are discrete samples of the continuous DFT in part c.

$$H[k] = H(\Delta f k) = H\left(\frac{W}{N} k\right)$$

$\hookrightarrow H[k]$ are sampled uniformly from $H(f)$ with intervals $\Delta f = \frac{W}{N}$

Thus as N increases, intervals become smaller and discrete samples more accurately traces out the shape of $|H(f)|^2$

2) 2-tap OFDM channel with $h_0=1$, $h_1=\frac{1}{2}$ and $N=4$

↳ $\hat{x}_0 \in \{-1, +1\}$, $\hat{x}_1 \in \{-1, +1\}$, $\hat{x}_2 \in \{-2, +2\}$, $\hat{x}_3 \in \{-2, +2\}$

↳ $w[m] \sim \mathcal{N}(0, 1)$ iid.

2.a) For IDFT outputs to be real valued, we need Hermitian symmetry of DCM. Namely

$$D[k] = D^*[N-k \pmod{N}]$$

$$D[0] = D^*[0] \quad \text{and} \quad D[2] = D^*[2] \Rightarrow D[0], D[2] \text{ real}$$

$$D[1] = D^*[3] \Rightarrow D[1] \text{ and } D[3] \text{ are complex conjugates of each other.}$$

Then one can simply assign $\hat{x}[m]$ to $D[m]$ as

$\left. \begin{aligned} D[0] &= \hat{x}[0] \\ D[2] &= \hat{x}[1] \end{aligned} \right\} \text{Real}$	and	$\left. \begin{aligned} D[1] &= \hat{x}[2] + j\hat{x}[3] \\ D[3] &= \hat{x}[2] - j\hat{x}[3] \end{aligned} \right\} \begin{array}{l} \text{Complex} \\ \text{Conjugate} \end{array}$
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$$b) H[k] = \sum_{n=0}^{N-1} h[n] e^{-j\frac{2\pi}{N}nk}$$

(no scaling factor $1/\sqrt{N}$ required here since included IDFT of $D[k]$)

↳ We truncate $h[n]$ to length 4 since we'll take 4-point DFT $\Rightarrow h = [1, 0.5, 0, 0]$

$$H[k] = h[0] + h[1] e^{-j\frac{\pi}{2}k} = 1 + 0.5 (-j)^k$$

$$H[0] = 1.5, \quad H[1] = 1 - 0.5j, \quad H[2] = 0.5, \quad H[3] = 1 + 0.5j$$

Revisit Option a) $\Rightarrow$ This part is optional not required for HW.

Now, recalling water filling argument, we should put more power onto higher SNR channels for optimally distributing total signal power

$$\left. \begin{array}{l} |H[0]|^2 = 2.25 \\ |H[1]|^2 = 1.25 \\ |H[2]|^2 = 1.25 \\ |H[3]|^2 = 0.25 \end{array} \right\} \begin{array}{l} \rightarrow \text{Assign one of the strong symbols to channel 0} \\ \quad \hookrightarrow D[0] = \hat{X}[2] \quad (\text{we could've also chosen } \hat{X}[3]) \\ \rightarrow \text{Assign one of the strong symbols } (\hat{X}[0], \hat{X}[1]) \text{ to} \\ \quad \text{channel 2 w/ 2nd highest SNR.} \\ \quad \hookrightarrow D[2] = \hat{X}[3] \end{array}$$

Then remaining symbols go to

$$\begin{aligned} D[1] &= \hat{X}[0] + j\hat{X}[3] \\ D[3] &= \hat{X}[0] - j\hat{X}[3] \end{aligned}$$

* Note that classical convention is what is used in option b, though.

c) In the freq. domain for each channel

$$Y[k] = H[k] D[k] + W[k]$$

where $W[k]$ is the DFT of $w[k]$

$$W[k] = \sum_{m=0}^3 w[m] e^{-j\frac{\pi}{2}mk}$$

$\Rightarrow$ Note that DFT is defined without the normalization constant as in slides. IDFT has $1/N$ factor.

$$\bullet W[0] = (w[0] + w[1] + w[2] + w[3])$$

$\hookrightarrow W[0]$ is purely real.

$$\hookrightarrow \text{Var}[W[0]] = (\underbrace{\text{Var}[w[0]]}_1 + \dots + \text{Var}[w[3]]) = 4$$

$$\text{Re}\{W[0]\} \sim \mathcal{N}(0, 4) \xrightarrow{\text{SD}} 2, \quad \text{Im}\{W[0]\} = 0$$

- $W[1] = (w[0] - jw[1] - w[2] + jw[3])$

$$\hookrightarrow \operatorname{Re}\{W[1]\} = (w[0] - w[2]) \Rightarrow \operatorname{Var}(\operatorname{Re}\{W[1]\}) = (1+1) = 2$$

$$\hookrightarrow \operatorname{Im}\{W[1]\} = (w[3] - w[1]) \Rightarrow \operatorname{Var}(\operatorname{Im}\{W[1]\}) = (1+1) = 2$$

$$\operatorname{Re}\{W[1]\} \sim \mathcal{N}(0, 2), \quad \operatorname{Im}\{W[1]\} \sim \mathcal{N}(0, 2)$$

- $W[2] = (w[0] - w[1] + w[2] - w[3])$

$\hookrightarrow W[2]$ is purely real.

$$\hookrightarrow \operatorname{Var}[W[2]] = \left(\underbrace{\operatorname{Var}[w[0]]}_{1} + \dots + \operatorname{Var}[w[3]] \right) = 4$$

$$\operatorname{Re}\{W[2]\} \sim \mathcal{N}(0, 4), \quad \operatorname{Im}\{W[2]\} = 0$$

- $W[3] = (w[0] + jw[1] - w[2] - jw[3])$

$$\hookrightarrow \operatorname{Re}\{W[3]\} = (w[0] + w[2]) \Rightarrow \operatorname{Var}(\operatorname{Re}\{W[3]\}) = (1+1) = 2$$

$$\hookrightarrow \operatorname{Im}\{W[3]\} = (w[3] - w[1]) \Rightarrow \operatorname{Var}(\operatorname{Im}\{W[3]\}) = (1+1) = 2$$

$$\operatorname{Re}\{W[3]\} \sim \mathcal{N}(0, 2), \quad \operatorname{Im}\{W[3]\} \sim \mathcal{N}(0, 2)$$

2.c) For $\hat{x}[0]$ ($n=0$)

$$Y[0] = H[0] D[0] + W[0] \quad \text{where} \quad H[0] = 1.5$$

$$\frac{H^*[0]}{|H[0]|} Y[0] = \underbrace{|H[0]|}_{\hat{h}_0} \hat{x}[0] + \frac{H^*[0]}{|H[0]|} W[0]$$

$$\hat{y}[0] = \hat{h}_0 \hat{x}[0] + W[0] \quad \text{where} \quad \operatorname{Var}[W[0]] = 4 \quad \nearrow \in \mathbb{R}$$

* Noting that decision boundary is 0 for $\hat{x}[0] \in \{-1, 1\}$

$$P_e(\hat{x}[0]) = Q\left(\frac{1.5}{\sqrt{4}}\right) = \boxed{Q(0.75)}$$

* Note that $|H[0]| \hat{x}[0] = \pm 1.5$

For $\hat{x}[1] (n=1)$

$$Y[2] = H[2] D[2] + W[0]$$

$$\frac{H[2]}{|H[2]|} Y[2] = \underbrace{|H[2]|}_{\hat{h}_1} \hat{x}[1] + \underbrace{\frac{H^*[2]}{|H[2]|} W[0]}_{\hat{w}[1]}$$

$$\hat{y}[1] = \hat{h}_1 \hat{x}[1] + \hat{w}[1]$$

$$\text{where } \hat{h}_1 = |H[2]| = 0.5$$

$$\text{Var}[\hat{w}[1]] = \text{Var}[W[0]] = 4$$

$$P_e(\hat{x}[1]) = Q\left(\frac{0.5}{\sqrt{4}}\right) = \boxed{Q(0.25)}$$

* Again decision boundary is 0 with $\hat{x}[1] \in \{-1, +1\}$

For $\hat{x}[2] (n=2)$

$$Y[1] = H[1] D[1] + W[1] \Rightarrow \underbrace{\frac{H^*[1]}{|H[1]|} Y[1]}_{\hat{y}[2]} = \underbrace{|H[1]|}_{\hat{h}_2} \hat{x}[2] + \underbrace{\frac{H^*[1]}{|H[1]|} W[1]}_{\hat{w}[2]}$$

$$\hat{y}[2] = \hat{h}_2 \hat{x}[2] + \hat{w}[2], \quad H[1] = 1 - 0.5j \Rightarrow |H[1]| = \sqrt{1.25}$$

$$H^*[1] W[1] = (1 + 0.5j)(W_R[1] + jW_I[1]) = W_R[1] - 0.5W_I[1] + j(0.5W_R[1] + W_I[1])$$

$$\text{Then } \text{Var}(\hat{w}[2]) = \frac{2 + \frac{1}{4} \cdot 2}{(\sqrt{1.25})^2} = 2$$

$$\Rightarrow \text{Note that, now } \hat{x}_2[2] \in \{-2, +2\} \Rightarrow \text{Thus } P_e(\hat{x}[2]) = Q\left(\frac{2\sqrt{1.25}}{\sqrt{2}}\right) = \boxed{Q(\sqrt{2.5})}$$

For $\hat{x}[3] (n=3)$

$$\text{Similarly } \hat{y}[3] = h_3 \hat{x}[3] + \hat{w}[3] \quad \text{where}$$

$$h_3 = |H[1]|$$

$$\hat{w}[3] = \text{Im}\left\{\frac{H^*[1]}{|H[1]|} W[1]\right\} = \frac{0.5W_R[1] + W_I[1]}{\sqrt{1.25}}$$

$$\text{Var}(\hat{w}[3]) = \frac{0.5 + 2}{1.25} = 2 \quad \text{and } \hat{x}_3 \in \{-2, 2\}$$

$$P_e(\hat{x}[3]) = Q\left(\frac{2\sqrt{1.25}}{\sqrt{2}}\right) = \boxed{Q(2.5)} \quad \text{similarly}$$

$$3) \quad y[m] = x[m] + 0.8x[m-1] + 0.5x[m-2] + w[m]$$

where $x[m] = \pm 1$ and $w[m] \sim \mathcal{N}(0, \sigma^2)$

↳ Block length is $N=8$ and we start with $\hat{x}[0], \dots, \hat{x}[7]$

3.a) We again require IDFT of $D[k]$ to be real valued which mean $D[k]$ must be Hermitian symmetric

$$D[m] = D^*[8-m \pmod{8}]$$

$$\rightarrow D[0] = D^*[0], \quad D[4] = D^*[4] \Rightarrow D[0] \text{ \& } D[4] \text{ real}$$

$$\rightarrow \left. \begin{array}{l} D[1] = D^*[7] \\ D[2] = D^*[6] \\ D[3] = D^*[5] \end{array} \right\} \text{ Hermitian conjugates}$$

$$\Rightarrow \vec{D} = \begin{bmatrix} \hat{x}[0] \\ \hat{x}[2] + j\hat{x}[3] \\ \hat{x}[4] + j\hat{x}[5] \\ \hat{x}[6] + j\hat{x}[7] \\ \hat{x}[1] \\ \hat{x}[6] - j\hat{x}[7] \\ \hat{x}[4] - j\hat{x}[5] \\ \hat{x}[2] - j\hat{x}[3] \end{bmatrix}$$

3.b) Since channel response is $L=3$ taps, the length of cyclic extension is $L-1=2$. We should add a suffix (at the end) instead of beginning. Then we should repeat the sequence starting from $d[0], d[1], \dots$

$$\vec{x} = [d[0] \ d[1] \ d[2] \ d[3] \ d[4] \ d[5] \ d[6] \ d[7] \ \underbrace{d[0] \ d[1]}_{\text{suffix}}]^T$$

3.c) Recall that $y[m] = x[m] + 0.8x[m-1] + 0.5x[m-2] + w[m]$

$$\begin{aligned} y[0] &= d[0] + 0.8x[-1] + 0.5x[-2] + w[0] \\ y[1] &= d[1] + 0.8d[0] + 0.5x[-1] + w[1] \\ y[2] &= d[2] + 0.8d[1] + 0.5d[0] + w[2] \\ y[3] &= d[3] + 0.8d[2] + 0.5d[1] + w[3] \\ y[4] &= d[4] + 0.8d[3] + 0.5d[2] + w[4] \\ y[5] &= d[5] + 0.8d[4] + 0.5d[3] + w[5] \\ y[6] &= d[6] + 0.8d[5] + 0.5d[4] + w[6] \\ y[7] &= d[7] + 0.8d[6] + 0.5d[5] + w[7] \\ y[8] &= d[0] + 0.8d[7] + 0.5d[6] + w[8] \\ y[9] &= d[1] + 0.8d[0] + 0.5d[7] + w[9] \end{aligned} \quad \left. \vphantom{\begin{aligned} y[0] \\ y[1] \\ y[2] \\ y[3] \\ y[4] \\ y[5] \\ y[6] \\ y[7] \\ y[8] \\ y[9] \end{aligned}} \right\} \text{interference from previous block}$$

* $y[0]$ and $y[1]$ contains terms from previous block ($x[-1], x[-2]$) so we should ignore them

* Remaining symbols ($y[2] \dots y[9]$) correctly replicate a circular convolution where signal wraps around itself (look at $y[8], y[9]$)

3.d) If we used a prefix, wrap-around would occur at the beginning and there would be interference terms we'd ignore

$$\begin{aligned} \hat{y}[0] &= d[0] + 0.8d[7] + 0.5d[6] + \hat{w}[0] \Rightarrow y[8] \\ \hat{y}[1] &= d[1] + 0.8d[0] + 0.5d[7] + \hat{w}[1] \Rightarrow y[9] \\ \hat{y}[2] &= d[2] + 0.8d[1] + 0.5d[0] + \hat{w}[2] \Rightarrow y[2] \\ &\vdots \\ \hat{y}[7] &= d[7] + 0.8d[6] + 0.5d[5] + \hat{w}[7] \Rightarrow y[7] \end{aligned}$$

Thus $\vec{\hat{y}} = [y[8] \ y[9] \ y[2] \ y[3] \ y[4] \ y[5] \ y[6] \ y[7]]^T$

3. e) First recall that , $\vec{D} = \begin{bmatrix} \hat{x}[0] \\ \hat{x}[2] + j\hat{x}[3] \\ \hat{x}[4] + j\hat{x}[5] \\ \vdots \\ \hat{x}[1] \\ \hat{x}[N_c-2] + j\hat{x}[N_c-1] \\ \vdots \\ \hat{x}[2] - j\hat{x}[3] \end{bmatrix}$

For $\hat{x}[0]$ ($n=0 \Rightarrow k=0$)

$$\frac{H^*[0]}{|H[0]|} \hat{y}[0] = |H[0]| \underbrace{D[0]}_{\hat{x}[0]} + \frac{H^*[0]}{|H[0]|} W[0] \Leftrightarrow \hat{y}[0] = \hat{h}_0 \hat{x}[0] + W[0]$$

$$\frac{H^*[0]}{|H[0]|} = +1 \Rightarrow \hat{y}[0] = -1 \Rightarrow \hat{x}[0] = -1$$

For $\hat{x}[1]$ ($n=1 \Rightarrow k=4$)

$$\frac{H^*[4]}{|H[4]|} \hat{y}[4] = |H[4]| \underbrace{D[4]}_{\hat{x}[1]} + \frac{H^*[4]}{|H[4]|} W[4] \Leftrightarrow \hat{y}[1] = \hat{h}_1 \hat{x}[1] + \hat{w}[1]$$

$$\frac{H^*[4]}{|H[4]|} = 1 \Rightarrow \hat{y}[4] = \hat{y}[1] = 1 \Rightarrow \hat{x}[1] = 1$$

For $\hat{x}[2]$ and $\hat{x}[3]$ ($k=1$)

$$\operatorname{Re}\left\{ \frac{H^*[1]}{|H[1]|} \hat{y}[1] \right\} = |H[1]| \hat{x}[2] + \operatorname{Re}\left\{ \frac{H^*[1]}{|H[1]|} W[1] \right\} \Rightarrow \hat{y}[2] = \hat{h}_2 \hat{x}[2] + \hat{w}[2]$$

$$\operatorname{Im}\left\{ \frac{H^*[1]}{|H[1]|} \hat{y}[1] \right\} = |H[1]| \hat{x}[3] + \operatorname{Im}\left\{ \frac{H^*[1]}{|H[1]|} W[1] \right\} \Rightarrow \hat{y}[3] = \hat{h}_2 \hat{x}[3] + \hat{w}[3]$$

$$\frac{H^*[1]}{|H[1]|} \hat{y}[1] = \frac{(0.5536 + j0.3768)(1+j)}{0.67} = 0.263 + j1.39$$

$$\hat{y}[2] = 0.263 > 0 \Rightarrow \hat{x}_2[1] = 1$$

$$\hat{y}[3] = 1.39 > 0 \Rightarrow \hat{x}_3[1] = 1$$

* Note that we only care about if $\hat{y}[n]$ are larger/smaller than zero since decision boundary is at $x=0$

For $\hat{x}[4]$ and $\hat{x}[5]$ ($k=2$)

Similarly we will now consider $\hat{y}[2]$ and $\hat{h}[2]$ since $D[2] = \hat{x}_4 + j\hat{x}[5]$

$$\frac{H[2]^*}{|H[2]|} Y[2] = \frac{(1+j)(0.1768 - j0.2828)}{|H[2]| \rightarrow \text{ignore since positive}} = \frac{-0.1060 + j0.4596}{|H[2]|}$$

$$\hat{y}[4] < 0 \Rightarrow \hat{x}[4] = -1$$

$$\hat{y}[5] > 0 \Rightarrow \hat{x}[5] = +1$$

For $\hat{x}[6]$ and $\hat{x}[7]$ ($k=3$)

Similarly we will now consider $\hat{y}[3]$ and $\hat{h}[3]$ since $D[3] = \hat{x}_6 + j\hat{x}[7]$

$$\frac{H[3]^*}{|H[3]|} Y[3] = \frac{(1+j)(0.1536 + j0.0232)}{|H[3]| \rightarrow \text{ignore since positive}} = \frac{0.1304 + j0.1768}{|H[3]|}$$

$$\hat{y}[6] > 0 \Rightarrow \hat{x}[6] = +1$$

$$\hat{y}[7] > 0 \Rightarrow \hat{x}[7] = +1$$

ML symbols are $\hat{x}_{ML} = [-1 \ 1 \ 1 \ 1 \ -1 \ 1 \ 1 \ 1]^T$

$$4) \quad y[m] = x[m] + \frac{1}{2} x[m-1] + w[m]$$

↳ modulation maps bit 0 $\rightarrow -1$, bit 1 $\rightarrow +1$

4. a) Since this is a $L=2$ -tap channel smallest prefix we can choose is.

$$v = L - 1 = \boxed{1}$$

4. b)

Data vector is $\hat{x} = \begin{bmatrix} -1 \\ +1 \\ -1 \\ +1 \\ -1 \\ +1 \\ +1 \\ +1 \end{bmatrix}$

4. c) Similar to previous questions we should have Hermitian symmetric $\vec{D}$

$$\vec{D} = \begin{bmatrix} \hat{x}[0] \\ \hat{x}[2] + j\hat{x}[3] \\ \hat{x}[4] + j\hat{x}[5] \\ \hat{x}[6] + j\hat{x}[7] \\ \hat{x}[1] \\ \hat{x}[6] - j\hat{x}[7] \\ \hat{x}[4] - j\hat{x}[5] \\ \hat{x}[2] - j\hat{x}[3] \end{bmatrix} = \begin{bmatrix} -1 \\ -1+j \\ -1+j \\ +1+j \\ +1 \\ +1-j \\ -1-j \\ -1-j \end{bmatrix}$$

4. d) In order to obtain $d[n]$ we take the N -point IDFT of $D[k]$

$$d[n] = \frac{1}{N} \sum_{k=0}^{N-1} e^{j \frac{2\pi}{N} kn} D[k] = \frac{1}{8} \sum_{k=0}^7 D[k] e^{j \frac{\pi}{4} kn}$$

$\frac{\pi}{4} \quad \frac{7\pi}{4}$

↳ Note that scaling factor $1/N$ is chosen to match MATLAB's $\text{ifft}(\cdot)$ function's result. Scaling could be set to $1/\sqrt{N}$ for both fft and ifft as well. However if ifft 's scaling is $1/N$, fft 's scaling should be 1.

$$d[0] = \frac{1}{8} \sum_{k=0}^7 D[k] = \frac{1}{8} (-2) = -0.25$$

$$d[1] = \frac{1}{8} \left[(D[0] - D[4]) + \sum_{k=1}^3 \left(D[k] e^{j\frac{\pi}{4}k} + D^*[k] e^{-j\frac{\pi}{4}k} \right) \right]$$

$$= \frac{1}{8} \left[-2 + 2 \sum_{k=1}^3 \text{Re} \{ D[k] e^{j\frac{\pi}{4}k} \} \right]$$

$$= \frac{1}{8} \left[-2 + 2 \text{Re} \{ \sqrt{2} e^{j\frac{3\pi}{4}} e^{j\frac{\pi}{4}} \} + 2 \text{Re} \{ \sqrt{2} e^{j\frac{3\pi}{4}} e^{j\frac{\pi}{2}} \} + 2 \text{Re} \{ \sqrt{2} e^{j\frac{\pi}{4}} e^{j\frac{3\pi}{4}} \} \right]$$

$$= \frac{1}{8} \left[-2 - 2\sqrt{2} + 2 - 2\sqrt{2} \right] = -0.5 - 0.5\sqrt{2} = -1.207$$

$$d[2] = \frac{1}{8} \left[D[0] + j D[1] - D[2] - j D[3] + D[4] + j D[5] - D[6] - j D[7] \right]$$

$$= \frac{1}{8} \left[1 - 1 + j(2j) - (-2) - j(2j) \right] = 0.25$$

$$d[3] = \frac{1}{8} \left[D[0] - D[4] + \sum_{k=1}^3 \left(D[k] e^{j\frac{3\pi}{4}k} + D^*[k] e^{-j\frac{3\pi}{4}k} \right) \right]$$

$$= \frac{1}{8} \left[-1 - 1 + 2 \text{Re} \{ \sqrt{2} e^{j\frac{3\pi}{4}} e^{j\frac{3\pi}{4}} \} + 2 \text{Re} \{ \sqrt{2} e^{j\frac{3\pi}{4}} e^{j\frac{6\pi}{4}} \} + 2 \text{Re} \{ \sqrt{2} e^{j\frac{\pi}{4}} e^{j\frac{3\pi}{4}} \} \right]$$

$$= \frac{1}{8} \left[-2 + 2\sqrt{2} \frac{1}{\sqrt{2}} \right] = 0$$

$$d[4] = -0.25, \quad d[5] = 0.207, \quad d[6] = 0.25, \quad d[7] = 0$$

↳ Rest is calculated by MATLAB's $\text{ifft}(\cdot)$

$$\vec{d} = \begin{bmatrix} -0.25 \\ -1.207 \\ 0.25 \\ 0 \\ -0.25 \\ 0.207 \\ 0.25 \\ 0 \end{bmatrix}$$

4.e) prefix $\leftarrow$

$$\vec{x} = \begin{bmatrix} d[7] \\ d[0] \\ d[1] \\ d[2] \\ d[3] \\ d[4] \\ d[5] \\ d[6] \\ d[7] \end{bmatrix} = \begin{bmatrix} 0 \\ -0.25 \\ -1.207 \\ 0.25 \\ 0 \\ -0.25 \\ 0.207 \\ 0.25 \\ 0 \end{bmatrix} \Rightarrow \text{New length is } N+L-1 = 9,$$

4.f) $y[m] = d[m] + \frac{1}{3} d[(m-1) \bmod 8]$

$$y[0] = d[0] + \frac{1}{3} d[7] = -0.25 + 0 = -0.25$$

$$y[1] = d[1] + \frac{1}{3} d[0] = -1.207 - \frac{0.25}{3} \approx -1.2904$$

⋮

```
% 1. Define the IDFT vector and the channel
d = [-0.25, -1.2071, 0.25, -0.5, -0.25, 0.2071, 0.25, 0];
h = [1, 1/3];
```

```
% 2. Add the cyclic prefix
x = [d(8), d];
```

```
% 3. Transmit through the channel using linear convolution
y_raw = conv(x, h);
```

```
% 4. Remove the prefix to extract the valid OFDM symbol
% y_raw(1) is the transient of the prefix.
% Indices 2 through 9 represent the 8 valid data symbols.
y = y_raw(2:9);
disp(y);
```

Using MATLAB's `conv()`

$$\vec{y} = [-0.25 \quad -1.2904 \quad -0.1524 \quad 0.0833 \quad -0.25 \quad 0.1238 \quad 0.3190 \quad 0.0833]^T$$

4.g) $Y[k] = \sum_{n=0}^7 y[n] e^{-j\frac{2\pi}{8}nk}$

Using MATLAB'S FFT command we find

```
Y = fft(y);
```

$$\vec{Y} = [-1.333 \quad -1+j1.414 \quad -0.6667+j1.333 \quad 1+j0.5286 \quad 0.6667 \quad 1-j0.5286 \quad -0.6667-j1.333 \quad -1-j1.414]^T$$

4.f) Note that without the noise $Y[k] = H[k] D[k]$

So, we can recover $D[k]$ with $D[k] = \frac{Y[k]}{H[k]}$

→ We again use MATLAB's fft command to compute DFT of channel coefficients ($\vec{h} = [1 \ 1/3]^T$) and compute $D[k]$

$$\vec{H} = \begin{bmatrix} 1.3333 + 0.0000i & 1.2357 - 0.2357i & 1.0000 - 0.3333i & 0.7643 - 0.2357i & 0.6667 + 0.0000i & 0.7643 + 0.2357i & 1.0000 + 0.3333i & 1.2357 + 0.2357i \end{bmatrix}^T$$

```
%% 5. Receiver DFT and Equalization (Part g)
```

```
Y = fft(y);  
H = fft(h, Nc);  
D_tilde = Y ./ H;
```

$$\vec{D} = [-1 \quad -1+i \quad -1+i \quad 1+i \quad 1 \quad 1-i \quad -1-i \quad -1-i]^T$$

$$\hat{x}[0] = D[0] = -1$$

$$\hat{x}[1] = D[4] = +1$$

$$\hat{x}[2] = \text{Re}\{D[1]\} = -1$$

$$\hat{x}[3] = \text{Im}\{D[1]\} = +1$$

$$\hat{x}[4] = \text{Re}\{D[2]\} = -1$$

$$\hat{x}[5] = \text{Im}\{D[2]\} = +1$$

$$\hat{x}[6] = \text{Re}\{D[3]\} = +1$$

$$\hat{x}[7] = \text{Im}\{D[3]\} = +1$$

Recovered bit sequence:

$$\vec{b}_{\text{recovered}} = [0 \ 1 \ 0 \ 1 \ 0 \ 1 \ 1 \ 1]^T$$

★ matches the original transmission vector